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The Editors
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Dear Editor,

I am pleased to submit my manuscript, “**Neighbourhood Structures and Ranking Operators for the Interval Job Shop Scheduling Problem**,” for consideration as a regular research article in *Applied Soft Computing*. The manuscript is transferred from *Computers & Operations Research*, where the handling editor assessed the results as interesting but considered the topic a better fit for a venue closer to the soft computing community — which is, indeed, where this work belongs: it extends a line of research on fuzzy and interval job shop scheduling developed largely in this journal and its sister venues, and its solver builds on artificial-bee-colony metaheuristics previously published in this field.

The paper addresses the Interval JSP, where processing times are bounded but uncertain, and contributes on both theoretical and computational fronts. On the theoretical side, I formalise the notion of *extreme critical paths* — arcs critical in either the best- or worst-case scenario graph — and define a neighbourhood $H(\sigma)$ on this basis, proving feasibility, connectivity and the no-loss-of-improving-neighbours property under any admissible interval ranking operator. Five neighbourhood variants (N_1 , N_2 , N_3 , N_{ext} and a new variant N_8) are formulated within this framework. On the empirical side, the paper reports a 12,300-run study on 82 benchmark instances of up to 1000 operations, organised in three phases: a 5×4 comparison of four interval ranking operators (EV, LEX1, LEX2, YX), a head-to-head comparison of the five neighbourhoods after per-neighbourhood irace tuning of a tabu search metaheuristic, and a comparison against three published IJSP solvers (GA, *fEABC*, ESABC). The irace-tuned tabu search with N_2 reduces mean relative error by $\approx 57\%$ over the previous best published solver on every one of the 82 instances.

Relative to the version examined at *Computers & Operations Research*, this revision makes the operational semantics of the interval makespan explicit: a new lemma shows that the interval computed by propagation is exactly the range of realizable makespans — its bounds are the best- and worst-case makespans — so the LEX2 ranking corresponds to lexicographic worst-case (robust) optimisation. A new related-work paragraph positions the approach with respect to the min–max regret literature on interval data and explains why regret-based criteria do not scale to job shop instances of this size.

In line with the journal’s emphasis on reproducibility, all raw run logs, aggregated CSVs, irace tuning records, instance files and analysis scripts are openly available on Zenodo ([10.5281/zenodo.20562888](https://zenodo.org/record/20562888)), with per-instance results on the 70 Taillard instances in a separate supplementary document ([10.5281/zenodo.20562857](https://zenodo.org/record/20562857)). I confirm that this manuscript has not been published previously, is not under consideration by any other journal, and that all listed contributors have approved the submission. I declare no competing financial interests or personal relationships that could have appeared to influence the work reported. Thank you for considering this submission.

Sincerely,

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